

HealthConnect Clinic Experience Lab

Week 6 — Analytics & Insights Advanced Analytics, Decision Support & Cross-Track Integration

Project: Improving Patient Appointment Attendance and Healthcare Support Using Data and AI

Track: Analytics & Insights (Data Analytics)

1. Week 6 Progress Summary

Week 6 extended the Week 5 analysis rather than repeating the initial EDA. The focus was deeper segmentation, validation of important findings, decision support, and preparation of analytical inputs for Data Science.

- Investigated previous no-show history with reminder status.
- Investigated booking lead time with reminder status.
- Investigated appointment type with previous no-show history.
- Validated the reminder/no-show relationship.
- Prepared cross-track inputs, recommendations, limitations, and Week 7 testing requirements.

2. Advanced Analysis & Validated Findings

Finding 1 — Previous No-Show History and Reminder Status

Patients with 0 previous no-shows had a current no-show rate of 43.51%, compared with 55.41% for patients with 1+ previous no-shows. Among the 1+ group, the no-show rate was 59.55% without a reminder and 53.83% with a reminder, a 5.72 percentage-point difference.

Business implication: Previous no-show history can help identify higher-risk patients for targeted appointment-support interventions.

Finding 2 — Booking Lead Time and Reminder Status

Lead time	No reminder	Reminder	Total
0–7 days	29.94%	27.06%	27.81%
8–14 days	34.78%	33.11%	33.55%
15–30 days	46.56%	41.88%	43.21%
31–60 days	63.64%	59.32%	60.49%

The overall no-show rate increased from 27.81% for 0–7 days to 60.49% for 31–60 days. The reminder group had a lower no-show rate in every lead-time group shown.

Business implication: Appointments booked further in advance may require earlier and/or additional reminder strategies.

Finding 3 — Appointment Type and Previous No-Show History

Appointment type	0 previous	1+ previous	Increase
Diagnostic Test	44.67%	56.47%	+11.80 pp
Follow-up	47.29%	56.78%	+9.49 pp
General Consultation	40.43%	55.26%	+14.83 pp
Specialist Consultation	43.89%	52.78%	+8.89 pp

Previous no-show history was associated with higher current no-show rates across all appointment types. General Consultation had the largest increase, 14.83 percentage points. Among patients with 1+ previous no-shows, Follow-up had the highest rate at 56.78%.

Finding 4 — Validated Reminder Relationship

Appointments with reminders had a no-show rate of 47.36%, compared with 51.39% without reminders, a 3.03 percentage-point difference. Lower no-show rates with reminders were also observed within the previous-no-show and booking-lead-time segments examined.

Interpretation: Reminder status is consistently associated with lower no-show rates in the analysed segments; this does not establish causation.

3. Data Science Integration

Receiving track: Data Science

Input: Validated risk patterns and potential interaction effects

Activity: Deeper segmentation and KPI validation

Expected use: Inform feature selection and investigation of interaction effects for no-show risk modelling.

Analytical finding	Evidence	Data Science use
Previous no-show history	43.51% vs 55.41%	Consider previous no-shows as a risk feature
Booking lead time	27.81% to 60.49%	Consider lead time as a risk-related feature
Reminder status	47.36% vs 51.39%	Include for further modelling investigation
Appointment type + history	Follow-up 56.78%; Diagnostic Test 56.47%	Investigate interaction effects among 1+

4. Business Recommendations & Actions

- Prioritize patients with previous no-shows for targeted appointment-support interventions.
- Consider earlier and/or additional reminders for appointments scheduled far in advance.
- Use targeted reminder strategies for higher-risk appointments.
- Give additional attention to high-risk combinations of previous no-show history and appointment type.
- Provide the identified analytical factors to Data Science to guide no-show risk-model development.

Overall recommendation: Move toward targeted appointment-support strategies based on patient history, booking lead time, reminder status, and appointment characteristics.

5. Limitations, Risks & Unresolved Issues

- Association does not establish causation.
- A recorded reminder does not confirm receipt, reading, or action by the patient.
- Smaller segments may produce less stable rates and require cautious interpretation.
- Available variables may not capture every factor influencing attendance.
- Cancellations and no-shows represent different outcomes and should remain distinct.

6. Week 7 Testing Requirements

- Test the consistency of the identified high-risk segments.
- Validate the risk factors within the developing predictive model.
- Test whether targeted reminder strategies can be implemented appropriately.
- Validate analytical findings against the final integrated solution.
- Assess whether recommendations can be translated into practical patient-support actions.

7. Week 6 Project Summary

Week 6 advanced Analytics & Insights from initial EDA and KPI calculation toward deeper decision-support analysis. The work identified consistent risk patterns involving previous no-show history, booking lead time, reminder status, and appointment type. These findings were translated into targeted business recommendations and prepared as analytical inputs for Data Science.

Key outcome: The analysis provides HealthConnect with a more targeted understanding of appointment no-show risk and establishes evidence for the next stage of modelling, testing, and end-to-end validation.